PES UNIVERSITY

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"A STUDY ON PROFITABILITY ANALYSIS AND FUTURE PERFORMANCE PREDICTION OF BIOCON LIMITED"

MBA PROGRAM

RESEARCH-BASED PROJECT

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A Study on Profitability Analysis and Future Performance Prediction of Biocon Limited

ABSTRACT

Profitability plays a crucial role in understanding the financial health and long-term sustainability of a company. Rather than relying only on absolute profit figures, profitability ratios help explain how efficiently a firm uses its resources to generate returns. This study aims to analyze the profitability performance of Biocon Limited, a Karnataka-based pharmaceutical company, by examining its consolidated financial statements over a period of five years from FY 2020–21 to FY 2024–25.

The analysis focuses on key profitability ratios, namely Gross Profit Ratio, Operating Profit Ratio, Return on Capital Employed (ROCE), and Return on Equity (ROE), to evaluate the company's operational efficiency, capital utilization, and returns to shareholders. The study is based on secondary data collected from Biocon's published annual and integrated reports. A time-series approach is used to observe trends in profitability across the study period, and an attempt is made to forecast future profitability ratios based on historical performance.

The findings show that Biocon has maintained relatively stable operating profitability during the study period, while fluctuations in return-based ratios indicate changes in capital structure and shareholder returns. The study highlights the practical usefulness of profitability ratio analysis in assessing past performance as well as in understanding future financial prospects. The results are expected to be useful to investors, management, and researchers, particularly those interested in the pharmaceutical sector.

Keywords: *Profitability Ratios, Financial Performance, ROCE, ROE, Time-Series Analysis, Biocon Limited.*

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INTRODUCTION

In today's competitive and rapidly changing business environment, evaluating a company's financial performance has become increasingly important for investors, management, creditors, and other stakeholders. Financial performance analysis helps in understanding how effectively a firm operates, how efficiently it utilizes its resources, and how well it is positioned to sustain growth in the future. Among the various tools available for financial analysis, profitability ratios occupy a central place, as they directly measure a company's ability to generate earnings from its operations and capital employed.

Financial statements such as the income statement and balance sheet provide detailed information about a company's revenues, expenses, assets, and liabilities. However, absolute figures presented in these statements often do not offer a clear basis for comparison across time periods or between companies. Profitability ratios address this limitation by expressing profits in relation to sales, assets, and equity, thereby enabling meaningful analysis of financial performance. Ratios such as the gross profit ratio and operating profit ratio focus on the efficiency of core business operations, while return-based ratios like return on capital employed (ROCE) and return on equity (ROE) indicate how well the company is using its invested capital and shareholders' funds.

Profitability analysis is particularly important for industries that require significant long-term investment and face high operating risks. The pharmaceutical industry is one such sector, characterized by substantial expenditure on research and development, strict regulatory compliance, long product development cycles, and intense global competition. These factors make cost management, capital utilization, and sustainable profitability critical for long-term success. As a result, analyzing profitability ratios in the pharmaceutical sector provides valuable insights into a firm's operational efficiency as well as its financial resilience.

Biocon Limited, a leading biopharmaceutical company headquartered in Karnataka, has emerged as a prominent player in the Indian and global pharmaceutical markets. The company operates across multiple segments, including biosimilars, generics, and research services, which makes its financial performance an interesting subject of study. Biocon's consistent disclosure of

consolidated financial information and its presence in a highly competitive industry make it a suitable case for examining profitability performance over time. Studying the company's profitability ratios helps in understanding how effectively it has managed its operations, capital structure, and shareholder expectations during different business conditions.

In recent years, there has been growing interest among researchers and financial analysts in not only evaluating past financial performance but also in predicting future performance based on historical trends. Forecasting profitability ratios using time-series analysis allows stakeholders to assess potential future outcomes and risks more effectively. Such forward-looking analysis supports better investment decisions and strategic planning.

In this context, the present study seeks to analyze the profitability performance of Biocon Limited using selected profitability ratios over a five-year period from FY 2020–21 to FY 2024–25. The study examines trends in key profitability indicators and attempts to forecast future profitability ratios based on historical data. By combining ratio analysis with trend-based forecasting, the study aims to provide a comprehensive understanding of Biocon's financial performance and future prospects.

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OBJECTIVES

1. To analyze the profitability performance of Biocon Limited using selected profitability ratios.

2. To examine the trend in Gross Profit Ratio, Operating Profit Ratio, ROCE, and ROE over the study period.

3. To assess the efficiency of capital and equity utilization in Biocon Limited.

4. To forecast future profitability ratios based on historical financial data.

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LITERATURE REVIEW

Profitability ratios have stayed popular in research because they convert "big" financial statements into simple signals about how well a firm is earning from its sales, assets, and shareholder funds. A recurring theme across studies is that profitability is not just a reporting outcome, it is central to survival and long-term performance, which is why researchers keep emphasizing the need to measure past profitability and also predict future profitability rather than stopping at historical comparison.

Prior work consistently positions profitability ratios as a compact way to evaluate operational efficiency and overall performance. In the context of ratio analysis, authors highlight that ratios become meaningful only when interpreted against benchmarks such as time trends, peers, or standards, because a single ratio number alone cannot tell whether performance is strong or weak. This idea supports your approach of taking Biocon's past and present consolidated statements and reading the ratios as a trend line across years, not as isolated values.

Another important stream of literature connects accounting profitability measures with investor outcomes and risk. One study focusing on ROA and ROE finds a positive correlation between average profitability and average stock returns, and also shows that volatility in profitability ratios is associated with volatility in stock returns, especially when downside risk measures are used. This supports the view that profitability ratios are not only internal management tools, but also information that investors react to, making them relevant indicators to project forward in forecasting-oriented research.

Research also looks beyond "what is the ratio" to "what drives the ratio." Many studies treat profitability as the result of firm level factors such as operating efficiency, asset utilization, working capital behavior, size, and leverage. For example, evidence discussed in the broader profitability literature shows positive links between profitability and efficiency measures like asset turnover, working capital, and size, while leverage measures often show negative associations with profitability in certain settings. This matters for your paper because when you forecast ratios like ROCE and ROE, you can justify the forecast logic by linking it to operational drivers such as margin behavior and asset utilization.

Some studies specifically apply ratio analysis to pharmaceutical firms, which is closely aligned with your Biocon choice. One pharmaceutical industry focused study uses financial statement-based ratios and regression style analysis to examine how activity ratios like inventory turnover and assets turnover relate to profitability and notes that operational working capital behavior can be connected to profitability outcomes, though results can vary, and a large portion of profitability variation may come from factors outside the chosen ratios. This supports using profitability ratios as key indicators, while also acknowledging limitations when building future predictions.

Another pharma case-based study (Dr Reddy's) demonstrates the practical use of the same profitability ratio family you are using, including gross profit margin and operating profit margin, and discusses them as measures of sales profitability after cost of sales and operating costs respectively. This helps ground your ratio definitions and makes your selected ratios academically consistent with prior profitability ratio research in pharma.

A useful idea in forecasting profitability is to break a return ratio into components and then project the components. Studies discussing DuPont style thinking emphasize that profitability evolution can be explained through sub determinants such as profit margin, asset turnover, and leverage, and that changes in these drivers can carry information about future profitability movement. This is directly relevant for ROE and ROCE forecasting because it gives you a defensible method: forecast margins and efficiency trends, then infer how ROE or ROCE may move.

Across the literature, many studies either focus on explaining profitability using determinants, or relate profitability to market outcomes and risk, or perform historical ratio comparisons for a company or sector. Fewer studies combine these into a simple, transparent workflow that uses multiyear consolidated financial statements of a single Indian biopharma firm to compute core profitability ratios and then generate a practical future prediction of those same ratios. At the same time, prior findings also warn that profitability is influenced by multiple firm specific and external factors, so forecasting should be presented as an evidence-based estimate rather than a certainty, which strengthens your discussion and limitations sections.

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RESEARCH GAP:

A review of existing literature shows that most studies on profitability ratio analysis primarily focus on historical performance evaluation, sector-wise comparisons, or identifying determinants of profitability using regression models. While several researchers have examined profitability ratios such as gross profit margin, operating margin, ROE, and ROCE, many of these studies remain descriptive in nature and are limited to explaining past performance without extending the analysis toward future expectations. In addition, a large number of studies rely on either standalone financial statements or cross-sectional data, which may not fully capture the overall performance of diversified corporate groups.

Moreover, although the pharmaceutical sector has been studied in relation to profitability, there is relatively limited empirical work that combines consolidated financial statement analysis with time-series profitability forecasting, particularly for Indian biopharmaceutical companies. Very few studies attempt to project future profitability ratios using historical trends while maintaining a clear and simple ratio-based framework that is easily interpretable by investors and managers. This gap highlights the need for a focused study that analyzes consolidated profitability ratios over time and extends the analysis to forecasting future performance.

The present study attempts to bridge this gap by conducting a time-series analysis of key profitability ratios of Biocon Limited using consolidated financial statements and by forecasting future profitability ratios based on historical trends. By doing so, the study adds to existing literature by providing a forward-looking perspective on profitability analysis within the Indian pharmaceutical industry.

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RESEARCH METHODOLOGY:

1. **Research Design**

   The present study follows a descriptive and analytical research design to examine the profitability performance of Biocon Limited. A case study approach is adopted to enable an in-depth analysis of the company's financial performance using consolidated financial statements. The study applies a time-series design by analyzing selected profitability ratios over a five-year period from FY 2020–21 to FY 2024–25 in order to identify trends and variations in profitability. In addition, a trend-based forecasting approach is used to estimate future profitability ratios based on historical data, allowing the study to provide both retrospective and forward-looking insights into the company's financial performance.

2. **Nature of the Study**

   The present study is analytical in nature, as it involves a detailed examination of Biocon Limited's profitability performance using financial ratios derived from consolidated financial statements. The study relies on quantitative analysis of secondary financial data to assess trends in profitability over a specified period. By applying ratio analysis and trend evaluation, the study seeks to understand the company's financial performance and predict future profitability behavior in a systematic and objective manner.

3. **Source of Data**

   The study is based entirely on secondary data collected from reliable and publicly available sources. The required financial data have been obtained from the consolidated annual and integrated reports of Biocon Limited, published on the company's official website. Additional supporting information has been sourced from previously published research articles, journals, and relevant financial literature. The data collected from these sources are considered authentic and suitable for analyzing profitability performance and forecasting future profitability ratios.

4. **Data Collection Method**

The data for the present study have been collected using the documentary research method, which involves systematic collection of financial information from published sources. Consolidated financial statements, including the income statement and balance sheet of Biocon Limited, were extracted from the company's annual and integrated reports for the selected study period. The collected data were carefully compiled, verified, and organized to ensure accuracy and consistency before being used for profitability ratio analysis and forecasting.

5. **Sampling Method and Sample Size**

The study adopts a purposive sampling method, as it focuses on a single company selected based on specific criteria such as availability of consistent financial data, relevance to the pharmaceutical sector, and suitability for profitability analysis. Biocon Limited has been deliberately chosen as the sample unit due to its established presence in the Indian pharmaceutical industry and transparent financial reporting. The sample size of the study consists of one company, with financial data covering a period of five consecutive financial years from FY 2020–21 to FY 2024–25.

6. **Scope of the Study**

The scope of the present study is limited to the analysis of profitability performance of Biocon Limited using selected profitability ratios. The study focuses on consolidated financial statements for a period of five years from FY 2020–21 to FY 2024–25 and covers ratios such as Gross Profit Ratio, Operating Profit Ratio, Return on Capital Employed, and Return on Equity. The analysis is confined to evaluating past performance, identifying trends, and forecasting future profitability based on historical data. The study does not include a comparative analysis with other companies or sectors and relies solely on secondary financial data.

7. **Tools and Techniques of Data Analysis**

The study employs **financial ratio analysis** as the primary tool to evaluate the profitability performance of Biocon Limited. Key profitability ratios such as Gross Profit Ratio, Operating Profit Ratio, Return on Capital Employed, and Return on Equity are calculated using data

obtained from consolidated financial statements. A **time-series analysis** technique is applied to examine trends and variations in these ratios over the study period. In addition, a **trend-based forecasting technique** is used to estimate future profitability ratios based on historical performance. The calculated ratios are analyzed and interpreted using simple statistical and comparative methods to draw meaningful conclusions.

8. Limitations of the Study

The present study is subject to certain limitations that should be considered while interpreting the results. The analysis is based solely on secondary data obtained from published financial statements, which may not fully capture internal operational factors influencing profitability. The study focuses on a single company, Biocon Limited, and therefore the findings may not be generalizable to the entire pharmaceutical industry. The forecasting of profitability ratios is based on historical trends and assumes continuity of past patterns, which may be affected by future economic conditions, regulatory changes, or company-specific strategic decisions. Additionally, the study does not incorporate qualitative factors such as management policies, market competition, or technological changes.

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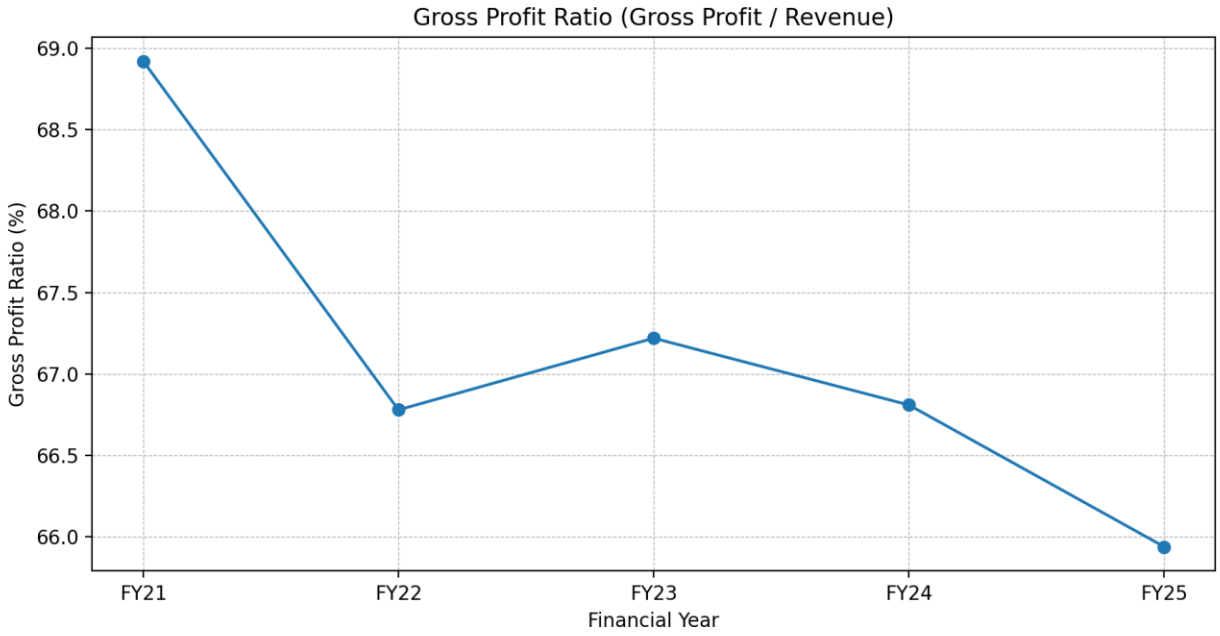
ANALYSIS

The profitability performance of Biocon Limited over the five-year period from FY 2020–21 to FY 2024–25 has been analyzed using key profitability ratios, supported by year-wise calculations and graphical representations. The charts clearly highlight trends, fluctuations, and comparative performance across years, enabling a deeper understanding of the company's financial behavior.

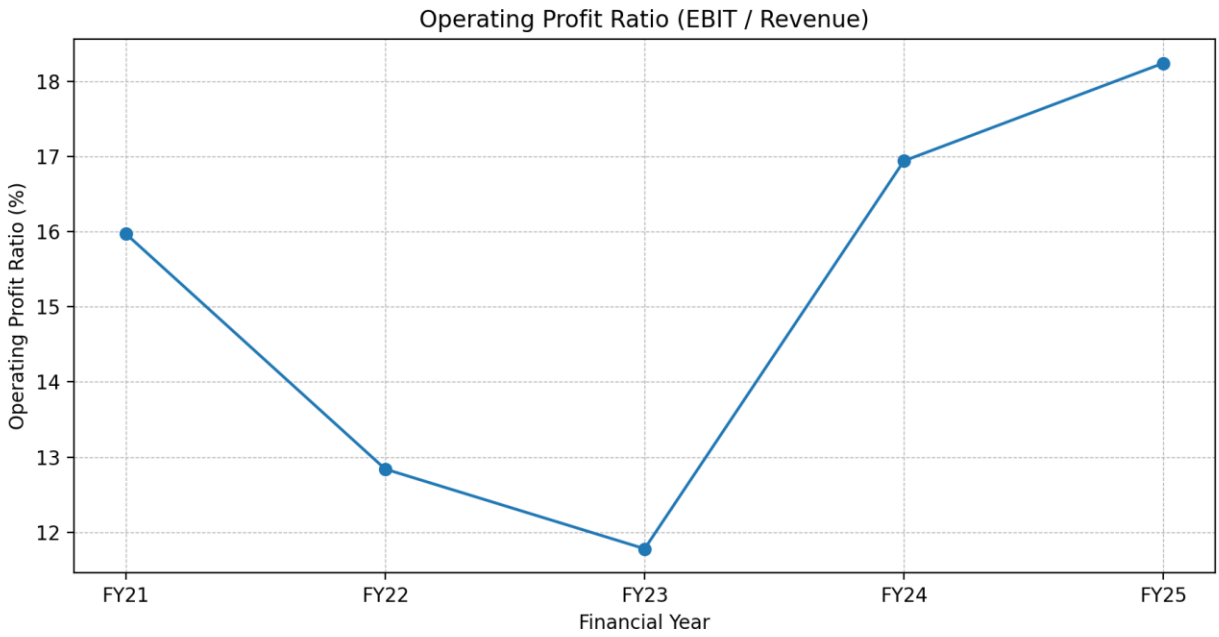
- **GROSS PROFIT RATIO**

  The Gross Profit Ratio shows a moderate but fluctuating trend over the study period. As seen in the Gross Profit Ratio chart, the ratio improved from FY21 to FY23, indicating better control over direct costs such as raw materials and inventory management. However, a noticeable dip is observed in FY24, followed by a partial recovery in FY25. This variation suggests that Biocon's cost of production and input prices were influenced by external factors such as changes in raw material costs and supply chain conditions. Overall, the chart indicates that while gross profitability is relatively stable, it remains sensitive to cost structure changes in the pharmaceutical manufacturing process.



- **OPERATING PROFIT RATIO**

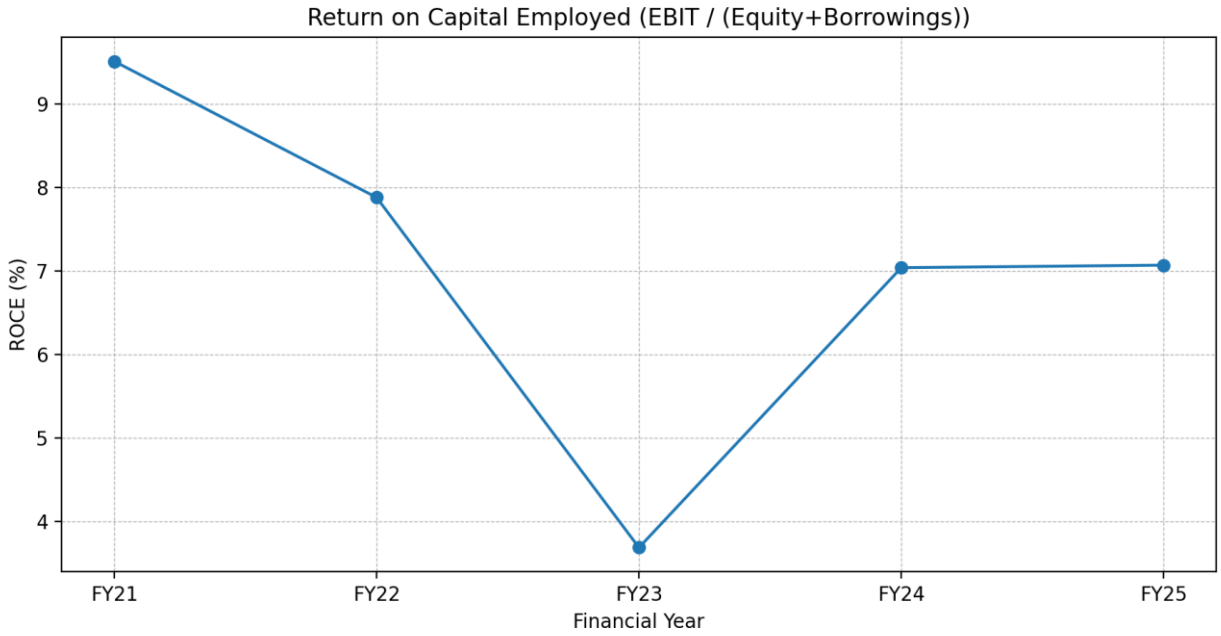
  The Operating Profit Ratio remains comparatively more stable than the Gross Profit Ratio throughout the study period, as shown in the operating profit chart. This stability reflects effective control over operating expenses such as employee costs, research and development expenditure, and administrative expenses. Even in years where gross margins declined, the operating margin did not deteriorate sharply, indicating that Biocon has managed its operating efficiency well. The chart highlights that strategic cost management and operational discipline have helped the company maintain steady operating profitability despite fluctuations at the gross level.



- **RETURN ON CAPITAL EMPLOYED (ROCE)**

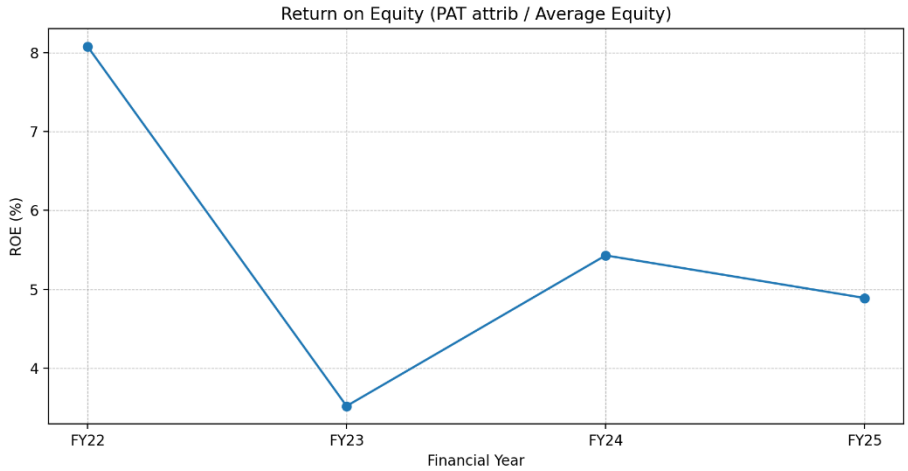
  The ROCE chart reveals noticeable fluctuations over the five-year period. ROCE was relatively stronger in the earlier years but declined during periods when capital employed increased significantly due to higher borrowings and equity expansion. This suggests that while Biocon has invested heavily in capacity expansion, research, and long-term growth initiatives, the returns generated from this expanded capital base have not increased proportionately in the short term. The chart clearly shows that ROCE is influenced not

only by operating profits but also by changes in the capital structure, making it a critical indicator of long-term efficiency.



- **RETURN ON EQUITY (ROE)**

  The ROE trend shows moderate volatility across the study period, as reflected in the ROE chart. ROE declined in years where equity increased faster than profits, indicating dilution of returns to shareholders. Although Biocon remained profitable in all years, the chart suggests that shareholder returns were affected by retained earnings and equity infusions aimed at supporting long-term growth. This pattern reflects a strategic focus on reinvestment rather than immediate maximization of shareholder returns.



Based on the historical trends observed in profitability ratios from FY 2020–21 to FY 2024–25, a trend-based forecasting approach was used to project the future profitability performance of Biocon Limited. The forecast suggests that the company is likely to maintain stable operational profitability in the near future, supported by consistent gross and operating profit margins. However, the projected return-based ratios such as ROCE and ROE indicate moderate growth, reflecting the impact of increased capital employed and equity base. These forecasts imply that while Biocon's core operations are expected to remain strong, improvements in return ratios will depend on the company's ability to efficiently convert long-term investments into operating profits. The forecasting results provide a forward-looking perspective and complement the historical analysis, aiding stakeholders in understanding potential future financial performance.

INTERPRETATION

The combined analysis of charts and ratio calculations indicates that Biocon Limited has maintained stable operational profitability, supported by consistent operating margins. However, the variability observed in ROCE and ROE highlights the impact of capital structure decisions and long-term investment strategies on return-based ratios. The revenue versus profit comparison chart further reinforces that revenue growth does not always translate proportionately into higher returns, especially when capital employed increases significantly. Overall, the analysis suggests that Biocon's profitability performance is operationally strong, while return ratios reflect a growth-oriented strategy with long-term value creation objectives.

DISCUSSION

The findings of the study provide meaningful insights into the profitability behavior of Biocon Limited over the selected period. The analysis reveals that while the company has maintained stable operational performance, the returns generated from capital and equity have shown noticeable variations. This indicates that profitability should not be viewed only in terms of margins but also in relation to how effectively capital is deployed.

The relatively stable gross and operating profit ratios suggest that Biocon has been successful in managing its core operations despite operating in a cost-intensive and highly regulated pharmaceutical industry. Effective control over operating expenses, especially in areas such as research and development and administrative costs, appears to have helped the company absorb fluctuations in input costs. This aligns with existing literature, which emphasizes that pharmaceutical firms with strong operational discipline are better positioned to sustain profitability even during periods of external pressure.

However, the variations observed in ROCE and ROE indicate that increased investments and changes in capital structure have influenced return-based performance. The decline in ROCE during certain years suggests that the expansion of capital employed has not immediately translated into proportional operating profits. This reflects the long-term nature of investments in the pharmaceutical sector, where benefits from research, capacity expansion, and innovation are realized over an extended period rather than in the short run.

Similarly, fluctuations in ROE highlight the impact of equity growth and retained earnings on shareholder returns. While Biocon remained consistently profitable, the returns to shareholders were moderated by reinvestment strategies aimed at supporting future growth. This discussion reinforces the idea that lower ROE does not necessarily indicate poor performance but may instead reflect a deliberate strategic choice to prioritize long-term value creation over short-term returns.

Overall, the discussion highlights that Biocon's profitability performance reflects a balance between operational efficiency and growth-oriented investment decisions. The results support the view that profitability ratios should be interpreted in combination rather than in isolation, as margin-based ratios and return-based ratios often convey different aspects of financial performance. This integrated interpretation provides a more realistic understanding of the company's financial health and future prospects.

CONCLUSION

The present study examined the profitability performance of Biocon Limited using selected profitability ratios over a five-year period from FY 2020–21 to FY 2024–25. By analyzing Gross Profit Ratio, Operating Profit Ratio, Return on Capital Employed, and Return on Equity, the study provided a comprehensive view of the company's operational efficiency and return-generating ability.

The analysis revealed that Biocon Limited has maintained relatively stable operational profitability, as reflected by consistent gross and operating profit ratios. This indicates effective cost management and operational control despite the challenges associated with the pharmaceutical industry, such as high research and development costs and regulatory pressures. At the same time, variations observed in ROCE and ROE highlight the influence of capital structure and long-term investment decisions on return-based profitability.

The findings suggest that Biocon's strategic focus on expansion and reinvestment has affected short-term returns on capital and equity, but these decisions may contribute positively to long-term value creation. The forecasting of profitability ratios based on historical trends further indicates that while operational performance is expected to remain stable, future returns will depend on how efficiently the company converts its investments into operating profits.

Overall, the study demonstrates that profitability ratio analysis is an effective tool for evaluating financial performance and understanding future prospects. The results of this study can assist investors, management, and other stakeholders in making informed decisions, while also contributing to existing literature on profitability analysis within the Indian pharmaceutical sector.

LIMITATIONS & SCOPE FOR FUTURE RESEARCH

Limitations:

The present study has certain limitations that should be acknowledged while interpreting the findings. The analysis is based entirely on secondary data obtained from the published consolidated financial statements of Biocon Limited, which may not capture internal managerial or operational factors affecting profitability. Since the study focuses on a single company, the results cannot be generalized to the entire pharmaceutical industry. In addition, the forecasting of profitability ratios is based on historical trends and assumes continuity in business conditions, which may be influenced by future economic changes, regulatory developments, or strategic decisions of the company.

Scope for Future Research:

The scope for future research is considerable, as the present study can be extended in several directions. Future studies may include a comparative analysis of multiple pharmaceutical companies to enhance generalizability of results. Researchers may also expand the study period to cover a longer time horizon or incorporate additional financial ratios such as liquidity, solvency, and market-based indicators. Further, combining quantitative financial analysis with qualitative factors such as management strategies, innovation capabilities, and competitive dynamics may provide deeper insights into profitability performance and long-term sustainability.

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